

Business Problem for Financial analysis dataset

Columns:

- Date
- Department
- Revenue
- Expenses
- Profit

A company wants to evaluate the **financial performance of its departments** (Marketing, IT, Operations, Finance) to understand how effectively revenue is being generated and controlled against expenses. Management needs a dashboard to monitor profitability trends, identify departments with high costs, and detect periods of financial loss.

The objective is to support **data-driven decision making** by analyzing revenue, expenses, and profit across departments and over time. The dashboard should help management answer the following questions:

- Which departments generate the highest revenue?
- Which departments incur the highest expenses?
- Which departments are consistently profitable or operating at a loss?
- How does profit change over time?
- Are there periods where expenses exceed revenue?

Key Performance Indicators (KPIs)

Create the following KPIs in Power BI using measures.

1. Total Revenue

Total revenue generated by all departments.

2. Total Expenses

Total operational expenses across departments.

3. Total Profit

Overall profitability of the organization.

4. Profit Margin

Measures profitability relative to revenue.

Chart Requirements for the Dashboard

1. KPI Cards

Visual Type: Card

Display the following metrics:

- Total Revenue
- Total Expenses
- Total Profit
- Profit Margin

Purpose: Provide an overview of the organization's financial performance.

2. Revenue by Department

Visual Type: Clustered Column Chart

Axis: Department

Values: Sum of Revenue

Purpose: Identify which department generates the most revenue.

3. Expenses by Department

Visual Type: Bar Chart

Axis: Department

Values: Sum of Expenses

Purpose: Determine which departments incur the highest costs.

4. Profit by Department

Visual Type: Column Chart

Axis: Department

Values: Sum of Profit

Purpose: Compare profitability across departments.

5. Financial Trend Over Time

Visual Type: Line Chart

Axis: Date

Values:

- Sum of Revenue
- Sum of Expenses
- Sum of Profit

Purpose: Track financial performance trends over time.

6. Treemap

Purpose: Show proportional contribution of departments.

Configuration:

Group: Department

Values: Revenue or Profit

Insight:

Helps identify which department contributes the most financially.

Treemap works well when you want to show proportion rather than exact comparison.

7. Department Profit Distribution

Visual Type: Donut Chart

Legend: Department

Values: Sum of Profit

Purpose: Show each department's contribution to total profit.

OUTPUT:



